Research Statement

My primary research interests lie in natural language processing (NLP), treatment effect estimation, and the application of machine learning to economic research. I am particularly interested in how emerging computational methods can expand the range of information available for economic analysis and how econometric methods can be developed and applied to identify the causal effects of policies. My research combines methodological development with empirical applications, with a particular emphasis on environmental economics and climate policy. Through my doctoral research and previous experience at the Korea Development Institute (KDI), I have developed a broader interest in integrating traditional econometric approaches with computational techniques to address economic questions that may be difficult to investigate using conventional data and methods.

One of my principal research interests is the application of NLP to economic questions. Economic research has traditionally used structured numerical data. However, substantial amounts of economically relevant information exist in unstructured formats, particularly text. Public discussions, policy announcements, corporate disclosures, and social media interactions contain information about individual preferences, expectations, and perceptions that may not be adequately captured by conventional datasets. Recent advances in artificial intelligence and machine learning have substantially improved researchers' ability to process and analyze these sources, creating new opportunities for empirical economic research.

My second major research interest concerns the identification and estimation of treatment effects. Policy evaluation frequently relies on observational data in which treatment assignment, eligibility, and participation are not necessarily random. I am interested in understanding how these features affect causal identification and how econometric frameworks can be extended to accommodate institutional complexities that are not fully represented in conventional specifications.

My research interests have been shaped by my experience as a researcher at KDI from 2021 to 2023. During this period, I provided research assistance for several published policy reports addressing carbon taxation, emissions trading systems, renewable energy procurement, and the transition toward a low-carbon economy. These included *A Study on the Introduction of a Carbon Tax* (2021), *Strategies and Policy Challenges for the Transition to a Low-Carbon Economy* (2023), and *Improving Renewable Energy Procurement Systems and Support Policies* (2023).

My contributions included collecting and organizing data used in policy analysis, and also performing preliminary analysis where requested. Working on these projects provided valuable exposure to the practical challenges of economic research, particularly the importance of institutional knowledge, data availability, and careful interpretation of policy mechanisms. The experience also strengthened my interest in environmental economics and helped motivate my subsequent doctoral research on climate-related public opinion and the evaluation of carbon-pricing policies.

Looking forward, I hope to develop a research agenda that increasingly integrates NLP, machine learning, and causal inference. I view machine learning not as a replacement for traditional econometric methods but as a complementary set of tools that can expand empirical possibilities, particularly when researchers encounter unstructured information, high-dimensional covariates, or heterogeneous economic relationships.

One direction I intend to pursue is the use of NLP to construct measures of public perceptions, expectations, and responses to economic policies. By combining text-based measures with conventional economic datasets, I hope to investigate how public attitudes relate to policy acceptance, economic behavior, and policy outcomes. Climate policy provides a particularly relevant setting, although these methods are applicable to a much broader range of economic questions.

A second direction is to incorporate machine learning into treatment effect estimation. Methods such as double/debiased machine learning and causal forests offer potentially useful approaches for handling high-dimensional data and exploring treatment effect heterogeneity. I am interested in examining how these techniques can complement established identification strategies, particularly in empirical settings where policy eligibility changes over time or where treatment effects vary across individuals, firms, or institutions. Importantly, I intend to maintain a clear distinction between improved predictive performance and credible causal identification.

Ultimately, my objective is to contribute to economic research by combining methodological rigor with an openness to emerging analytical techniques. I hope to develop and apply methods that make previously inaccessible information useful for empirical analysis while improving our ability to identify and interpret the effects of economic policies. Through the integration of NLP, machine learning, and treatment effect estimation, I aim to pursue research that contributes both to econometric methodology and to the empirical understanding of economic and policy-related questions.